Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 06)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

0 1 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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0 1 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

1 .....

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2 .....

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0 1 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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0 1 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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0 2 . 1 Explain why organic pollution of rivers is usually monitored by measuring Biochemical Oxygen Demand (BOD) rather than the concentrations of the pollutants themselves.

[3 marks]

0 2 . 2 Explain why a coliform count is used to assess the health risk from sewage contamination rather than testing for pathogens such as typhoid.

[2 marks]

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0 2 . 3 Give **three** characteristics of species that make them useful in a biotic index.

[3 marks]

1 .....

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2 .....

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3 .....

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0 2 . 4 Give **one** advantage and **one** disadvantage of using a biotic index rather than chemical monitoring.

[2 marks]

Advantage .....

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Disadvantage .....

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10

0 3 . 1 Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

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0 3 . 2 Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

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0 3 . 3 Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

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0 3 . 4 Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

1 .....

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2 .....

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10

0 4 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m^{-2} day^{-1}	Mean solar energy in December / kWh m^{-2} day^{-1}
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

0 4 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

Answer [ ] times

0 4 . 2 Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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0 4 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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0 4 . 4 Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

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0 4 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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0 5 . 1 Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

A Its total annual discharge ☐

B The albedo of its surface ☐

C Its salinity ☐

D The age of its bedrock ☐

0 5 . 2 Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

0 5 . 3 Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

5

0 6 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 6 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer _____ t/ha

0 6 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 6 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

1 .....

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2 .....

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0 7 . 1 Explain how each of the following features of vehicle design reduces energy use.

[6 marks]

Aerodynamics .....

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Reduced mass .....

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Regenerative braking (KERS) .....

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0 7 . 2 Explain why solid steel wheels on rails lose less energy than pneumatic tyres on roads.

[2 marks]

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0 7 . 3 Describe **two** ways combustion efficiency in an internal combustion engine can be improved.

[2 marks]

1 .....

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2 .....

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0 8 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 8 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 8 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 8 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 9 . 1 Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

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0 9 . 2 State what is meant by a **carbon footprint**.

[1 mark]

10

1 0 . 1 Discuss how the features of energy resources determine their suitability for different uses and their environmental sustainability.

[9 marks]

Extra space

1 0 . 2 State **one** reason why nuclear power stations can be located where transport infrastructure is poor.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS